

JEE (MAIN+ADVANCED)

Mathematics

ENTHUSIAST COURSE

$$\begin{matrix} & \begin{matrix} 1 & 2 & \dots & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ m \end{matrix} & \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{matrix}$$

STUDY MATERIAL

Matrices

English Medium

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MATRIX

1. INTRODUCTION :

A rectangular array of mn numbers (which may be **real or complex**) in the form of ' m ' horizontal lines (called **rows**) and ' n ' vertical lines (called **columns**), is called a matrix of order m by n , written as $m \times n$ matrix.

Such an array is enclosed by $[]$ or $()$ or $\|$. An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}$$

In compact form, the above matrix is represented by $A = [a_{ij}]_{m \times n}$. The number $a_{11}, a_{12}, \dots$ etc are known as the elements of the matrix A , a_{ij} belongs to the i^{th} row and j^{th} column and is called the **$(i, j)^{\text{th}}$ element** of the matrix $A = [a_{ij}]$.

e.g., $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 9 \end{bmatrix}$ is a matrix having 2 rows and 3 columns. Its order is 2×3 and it has 6 elements :

$$a_{11} = 1, a_{12} = 2, a_{13} = 3, a_{21} = 0, a_{22} = -1, a_{23} = 9.$$

2. SPECIAL TYPE OF MATRICES :

(a) **Row Matrix (Row vector)** : $A = [a_{11}, a_{12}, \dots, a_{1n}]$ i.e. row matrix has exactly one row.

(b) **Column Matrix (Column vector)** : $A = \begin{bmatrix} a_{11} \\ a_{12} \\ \vdots \\ a_{m1} \end{bmatrix}$ i.e. column matrix has exactly one column.

(c) **Zero or Null Matrix** : ($A = O_{m \times n}$) An $m \times n$ matrix whose all entries are zero.

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \text{ is a } 3 \times 2 \text{ null matrix \& } B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ is } 3 \times 3 \text{ null matrix}$$

(d) **Horizontal Matrix** : A matrix of order $m \times n$ is a horizontal matrix if $n > m$ e.g. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 1 & 1 \end{bmatrix}$

(e) **Vertical Matrix** : A matrix of order $m \times n$ is a vertical matrix if $m > n$ e.g. $\begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 6 \\ 2 & 4 \end{bmatrix}$

(f) **Square Matrix** : If number of rows = number of columns = n , then matrix is a square matrix of the order ' n '.

Note : The pair of elements a_{ij} & a_{ji} are called **Conjugate Elements**.

3. TRACE OF MATRIX :

The sum of the elements of a **square matrix** A lying along the principal diagonal is called the trace of A i.e.

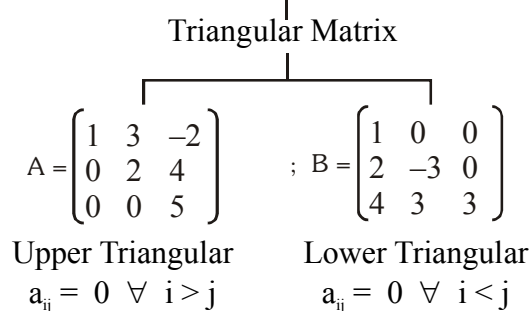
$$(\text{tr}(A)). \text{ Thus, if } A = [a_{ij}]_{n \times n}, \text{ then } \text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$$

Properties of trace of a matrix :

Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ and λ be a scalar then

$$(i) \quad \text{tr}(\lambda A) = \lambda \text{tr}(A) \quad (ii) \quad \text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$$

4. SQUARE MATRICES



Diagonal Matrix denoted as

$$A = \text{diag} (a_{11}, a_{22}, \dots, a_{nn})$$

where $a_{ij} = 0$ for $i \neq j$

Scalar Matrix

$$(a_{11} = a_{22} = a_{33} = a)$$

$$\begin{bmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & a \end{bmatrix}$$

Unit or Identity Matrix I

$$a_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Note :

- (i) Minimum number of zeros in triangular matrix of order n is $n(n-1)/2$.
- (ii) Minimum number of zero in a diagonal matrix of order n is $n(n-1)$.

5. EQUALITY OF MATRICES :

Let $A = [a_{ij}]$ & $B = [b_{ij}]$ are equal if

- (a) both have the same order.
- (b) $a_{ij} = b_{ij}$ for each pair of i & j .

Illustration 1 : Find the value of x, y, z and w which satisfy the matrix equation

$$\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4w-8 \end{bmatrix} = \begin{bmatrix} -x-1 & 0 \\ 3 & 2w \end{bmatrix}$$

Solution : As the given matrices are equal so their corresponding elements are equal.

$$x + 3 = -x - 1 \quad \Rightarrow \quad 2x = -4$$

$$\therefore x = -2 \quad \dots\dots\dots (i)$$

$$2y + x = 0 \quad \Rightarrow \quad 2y - 2 = 0 \quad [\text{from (i)}]$$

$$\Rightarrow y = 1 \quad \dots\dots\dots(\text{ii})$$

$$z - 1 = 3 \quad \Rightarrow \quad z = 4 \quad \dots\dots\dots(\text{iii})$$

$$4w - 8 = 2w \Rightarrow 2w = 8$$

$$\therefore w = 4 \quad \dots\dots\dots(\text{iv})$$

Ans.

6. ALGEBRA OF MATRICES :

Addition : $A + B = [a_{ij} + b_{ij}]$ where A & B are of the same order.

- (a) **Addition of matrices is commutative** : i.e. $A + B = B + A$
- (b) **Matrix addition is associative** : $(A + B) + C = A + (B + C)$
- (c) **Additive inverse** : If $A + B = \mathbf{O} = B + A$, then B is called **additive inverse** of A.
- (d) **Existence of additive identity** : Let $A = [a_{ij}]$ be an $m \times n$ matrix and $\mathbf{O}$ be an $m \times n$ zero matrix, then $A + \mathbf{O} = \mathbf{O} + A = A$. In other words, $\mathbf{O}$ is the **additive identity** for matrix addition.
- (e) **Cancellation laws** hold good in case of addition of matrices. If A,B,C are matrices of the same order, then $A + B = A + C \Rightarrow B = C$ (**left cancellation law**) and $B + A = C + A \Rightarrow B = C$ (**right-cancellation law**)

Note : The zero matrix plays the same role in matrix addition as the number zero does in addition of numbers.

Illustration 2 : If $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ & $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ and $A + B - D = \mathbf{O}$ (zero matrix), then D matrix will be-

(A) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 6 & 5 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$

$$(C) \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$$

$$(D) \begin{bmatrix} 0 & -2 \\ -3 & -7 \\ -5 & -6 \end{bmatrix}$$

Solution :

Let $D = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$$\therefore \mathbf{A} + \mathbf{B} - \mathbf{D} = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \begin{bmatrix} 1-1-a & 3-2-b \\ 3+0-c & 2+5-d \\ 2+3-e & 5+1-f \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{array}{l} -a=0 \Rightarrow a=0, \quad 1-b=0 \Rightarrow b=1, \\ 3-c=0 \Rightarrow c=3, \quad 7-d=0 \Rightarrow d=7, \\ 5-e=0 \Rightarrow e=5, \quad 6-f=0 \Rightarrow f=6 \end{array}$$

$$\therefore D = \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$$

Ans. (C)

Do yourself -1 :

- Find 2×3 matrix $[a_{ij}]_{2 \times 3}$, where $a_{ij} = i + 2j$
- Find the minimum number of zeroes in a lower triangular matrix of order 4.
- Find minimum number of zeros in a Scalar matrix of order 6.
- If $\begin{bmatrix} 2x+y & 2 & x-2y \\ a-b & 2a+b & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 4 & -1 & -3 \end{bmatrix}$, then find the values of x, y, a and b .
- Let $A = \begin{bmatrix} \alpha^2(a-2\lambda) & 0 & 3 \\ -1 & \alpha(b-3\lambda) & 1 \\ 0 & 5 & (c-4\lambda) \end{bmatrix}$; $B = \begin{bmatrix} \beta^2(a-2\lambda) & -5 & -3 \\ 7 & \beta(b-3\lambda) & 7 \\ 8 & 9 & (c-4\lambda) \end{bmatrix}$ and $C = \begin{bmatrix} \gamma^2(a-2\lambda) & -1 & 3 \\ 0 & \gamma(b-3\lambda) & 7 \\ 8 & -3 & (c-4\lambda) \end{bmatrix}$, where α, β, γ are real & distinct such that $\text{tr}(A) = \text{tr}(B) = \text{tr}(C) = -1$, where $\lambda \in \mathbb{N}$, then minimum possible value of $a + b + c$ is
- If $A = \begin{bmatrix} 2 & 3 & 9 \\ 8 & -2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & -7 & 2 \\ 6 & 4 & 8 \end{bmatrix}$, then find a matrix C such that $A - B + C = \mathbf{O}$ and also find the order of the matrix C .
- If $A = \begin{bmatrix} 8 & 9 \\ 7/2 & 8 \\ 1 & -1 \end{bmatrix}$, then find the additive inverse of A and show that additive inverse of additive inverse will be the matrix itself.

7. MULTIPLICATION OF A MATRIX BY A SCALAR :

If $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$; $kA = \begin{bmatrix} ka & kb & kc \\ kb & kc & ka \\ kc & ka & kb \end{bmatrix}$

Properties of scalar multiplication :

- (a) If A and B are two matrices of the same order and 'k' be a scalar then $k(A + B) = kA + kB$.
- (b) If k_1 and k_2 are two scalars and 'A' is a matrix, then $(k_1 + k_2)A = k_1A + k_2A$.
- (c) If k_1 and k_2 are two scalars and 'A' is a matrix, then $(k_1 k_2)A = k_1(k_2 A) = k_2(k_1 A)$

8. MULTIPLICATION OF MATRICES (Row by Column) :

Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$, then the matrix multiplication AB is possible if and only if $n = p$ and matrices are said to be **conformable** for multiplication.

In the product AB, A is called pre-factor and B is called post factor.

$\Rightarrow$ AB is possible if and only if number of columns in pre-factor = number of rows in post-factor.

Let $A_{m \times n} = [a_{ij}]$ and $B_{n \times p} = [b_{ij}]$, then order of AB is $m \times p$ & $(AB)_{ij} = \sum_{r=1}^n a_{ir} b_{rj}$

$$\text{e.g. } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2 \times 3} \text{ and } B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix}_{3 \times 4}$$

Then order of AB is 2×4 .

$$(AB)_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = \sum_{r=1}^3 a_{1r}b_{r1}$$

$$(AB)_{23} = a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} = \sum_{r=1}^3 a_{2r}b_{r3}$$

$$\text{In general } (ab)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} = \sum_{r=1}^3 a_{ir}b_{rj}$$

Illustration 3 : If $[1 \times 2] \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = \mathbf{O}$, then the value of x is :-

- (A) -1 (B) 0 (C) 1 (D) 2

Solution : The LHS of the equation

$$= [2 \quad 4x + 9 \quad 2x + 5] \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = [2x + 4x + 9 - 2x - 5] = 4x + 4$$

$$\text{Thus } 4x + 4 = 0 \Rightarrow x = -1$$

Ans. (A)

Illustration 4 : If A, B are two matrices such that $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$, then find AB.

Solution : Given $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (i) & $A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$ (ii)

Adding (i) & (ii)

$$2A = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$$

Subtracting (ii) from (i)

$$2B = \begin{bmatrix} -2 & 0 \\ 4 & 4 \end{bmatrix} \Rightarrow B = \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix}$$

$$\text{Now } AB = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 4 & 4 \end{bmatrix}$$

Ans.

9. PROPERTIES OF MATRIX MULTIPLICATION :

(a) Matrix multiplication is not commutative : i.e. $AB \neq BA$

Here both AB & BA exist and also they are of the same type but $AB \neq BA$.

Example :

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \text{ \& } B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \text{ then } AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \quad BA = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow AB \neq BA \text{ (in general)}$$

(b) $AB = \mathbf{O} \nRightarrow A = \mathbf{O} \text{ or } B = \mathbf{O}$ (in general)

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \text{ \& } B = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \text{ then } AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Note :

If A and B are two matrices such that

(i) $AB = BA$ then A and B are said to commute

(ii) $AB = -BA$ then A and B are said to anticommute

(c) Matrix Multiplication Is Associative :

If A , B & C are conformable for the product AB & BC , then $(AB)C = A(BC)$

(d) Distributivity :

$$\begin{aligned} A(B+C) &= AB+AC \\ (A+B)C &= AC+BC \end{aligned} \quad \text{Provided } A, B \text{ \& } C \text{ are conformable for respective products}$$

(e) $\text{tr}(AB) = \text{tr}(BA)$

Illustration 5 : Let $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix}$ & $C = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix}$ be the matrices

then, prove that in matrix multiplication cancellation law does not hold.

Solution :

We have to show that $AB = AC$; though B is not equal to C .

$$\text{We have } AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

$$\text{Now, } AC = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

Here, $AB = AC$ though B is not equal to C . Thus cancellation law does not hold in general.

Do yourself - 2 :

1. If $A = \begin{bmatrix} 2 & 9 \\ -4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 6 & 8 \end{bmatrix}$ and $C = \begin{bmatrix} 9 & -7 \\ -2 & 4 \end{bmatrix}$, then show that $A(B + C) = AB + AC$.

2. If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -1 & 1 \end{bmatrix}$, then prove that $(A - B)^2 \neq A^2 - 2AB + B^2$.

3. Find the value of x : $2 \begin{bmatrix} 3 & 1 & -2 \\ -1 & -3 & 4 \end{bmatrix} + x \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -2 \\ -8 & -14 & -2 \end{bmatrix}$

4. Let $A = \begin{bmatrix} -3 & 0 & 2 \\ 1 & x & 5 \\ -2 & 0 & x^2 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ b \\ -1 \end{bmatrix}$ and $C = [3 \ 5 \ 1]$, then the number of integral value(s) of 'b' for

which $\text{Tr}(ABC) \leq -18 \forall x \in \mathbb{R}$ is (are)

5. The number of values of n such that at least one pair of two matrices (A, B) of order n exists such that $AB - BA = I$, is

10. POSITIVE INTEGRAL POWERS OF A SQUARE MATRIX :

For a square matrix A , $A^n = \underbrace{A \cdot A \cdot A \dots A}_{\text{upto } n \text{ times}}$, where $n \in \mathbb{N}$

Note :

(i) $A^m \cdot A^n = A^{m+n}$

(ii) $(A^m)^n = A^{mn}$, where $m, n \in \mathbb{N}$

(iii) If A and B are square matrices of same order and $AB = BA$ then

$$(A + B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + {}^nC_2 A^{n-2} B^2 + \dots + {}^nC_n B^n$$

Note that for a unit matrix I of any order, $I^m = I$ for all $m \in \mathbb{N}$.

Do yourself - 3 :

1. If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then using principle of mathematical induction or otherwise prove that

$$A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix} \text{ where } n \text{ is a positive integer.}$$

2. If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, where $i = \sqrt{-1}$ and $x \in \mathbb{N}$, then A^{4x} equals -

(A) $\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$ (B) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$

3. For $\theta = \frac{3\pi}{5}$, let $B = [b_{ij}]$ be a square matrix of 2^{nd} order such that

$$b_{ij} = \begin{cases} \cos \theta & , i = j \\ \cos\left(\frac{j\pi}{2} + \theta\right) & , i > j \\ \sin\left(\frac{j\pi}{2} - \theta\right) & , i < j \end{cases} \text{ then trace of matrix } B^5 \text{ is equal to -}$$

(A) -2 (B) -1 (C) 1 (D) 2

4. If A and B are 3×3 matrices such that $AB = A$ and $BA = B$, then-

(A) $A^2 = A, B^2 \neq B$ (B) $A^2 \neq A, B^2 = B$
(C) $A^2 = A, B^2 = B$ (D) $A^2 \neq A, B^2 \neq B$

5. If the matrix A is given by $A = \begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$ then determinant value of $(A^{27} - 2A^{26})$ is

(A) -5 (B) -7 (C) -10 (D) 0

6. If A, B are two square matrices such that $AB = A, BA = B$ then $(A + B)^7$ equals

(A) $A + B$ (B) $2^7(A + B)$ (C) $2^6(A + B)$ (D) $2^8(A + B)$

7. A square matrix P satisfies $P^2 = I - P$, where I is the identity matrix. If $P^n = 5I - 8P$ ($n \in \mathbb{N}$), then minimum value of n is equal to -

(A) 4 (B) 5 (C) 6 (D) 7

8. Let A be a square matrix satisfying $A^2 = 2I$ and $\sum_{k=1}^{2n} A^k = 63A(A + I)$, then the value of n is (where I denotes identity matrix)

11. SPECIAL SQUARE MATRICES :

- (a) **Idempotent Matrix** : A square matrix is idempotent provided $A^2 = A$.

For idempotent matrix note the following :

- (i) $A^n = A \quad \forall n \geq 2, n \in \mathbb{N}$.
- (ii) determinant value of idempotent matrix is either 0 or 1

- (b) **Periodic Matrix** : A square matrix which satisfies the relation $A^{k+1} = A$, for some positive integer K , is a periodic matrix. The period of the matrix is the least value of K for which this holds true.

Note that period of an idempotent matrix is 1.

- (c) **Nilpotent Matrix** : A square matrix of the order 'n' is said to be nilpotent matrix of order m , $m \in \mathbb{N}$, if $A^m = \mathbf{O}$ & $A^{m-1} \neq \mathbf{O}$.

- (d) **Involutory Matrix** : If $A^2 = I$, the matrix is said to be an involutory matrix. i.e. square roots of identity matrix is involutory matrix.

Note : The determinant value of involutory matrix is 1 or -1.

Illustration 6 : Show that the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent.

Solution :

$$A^2 = A.A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} \times \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 2.2 + (-2).(-1) + (-4).1 & 2(-2) + (-2).3 + (-4).(-2) & 2.(-4) + (-2).4 + (-4).(-3) \\ (-1).2 + 3.(-1) + 4.1 & (-1).(-2) + 3.3 + 4.(-2) & (-1).(-4) + 3.4 + 4.(-3) \\ 1.2 + (-2).(-1) + (-3).1 & 1.(-2) + (-2).3 + (-3).(-2) & 1.(-4) + (-2).4 + (-3).(-3) \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = A$$

Hence the matrix A is idempotent.

Illustration 7 : Show that $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is nilpotent matrix of order 3.

Solution : Let $A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$

$$A^2 = A.A = \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 1+5-6 & 1+2-3 & 3+6-9 \\ 5+10-12 & 5+4-6 & 15+12-18 \\ -2-5+6 & -2-2+3 & -6-6+9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix}$$

$$A^3 = A^2.A = \begin{bmatrix} 0 & 0 & 0 \\ 3 & 3 & 9 \\ -1 & -1 & -3 \end{bmatrix} \times \begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 \\ 3+15-18 & 3+6-9 & 9+18-37 \\ -1-5+6 & -1-2+3 & -3-6+9 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{O}$$

$$\therefore A^3 = \mathbf{O} \text{ i.e., } A^k = \mathbf{O}$$

Here $k = 3$

Hence A is nilpotent of order 3.

Illustration 8 : Show that the matrix $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is involutory.

Solution :

$$A^2 = A.A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \times \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 25-24+0 & 40-40+0 & 0+0+0 \\ -15+15+0 & -24+25+0 & 0+0+0 \\ -5+6-1 & -8+10-2 & 0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

Hence the given matrix A is involutory.

Illustration 9 : Show that a square matrix A is involutory, iff $(I - A)(I + A) = \mathbf{O}$

Solution : Let A be involutory

$$\text{Then } A^2 = I$$

$$(I - A)(I + A) = I^2 + IA - AI - A^2 = I + A - A - A^2 = I - A^2 = \mathbf{O}$$

Conversely, let $(I - A)(I + A) = \mathbf{O}$

$$\Rightarrow I^2 + IA - AI - A^2 = \mathbf{O} \Rightarrow I + A - A - A^2 = \mathbf{O}$$

$$\Rightarrow I - A^2 = \mathbf{O} \Rightarrow A \text{ is involutory}$$

Do yourself - 4 :

1. The matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$ is

- (A) idempotent matrix (B) involutory matrix
(C) nilpotent matrix (D) periodic matrix

2. If $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$ and A^2 is the identity matrix, then find the value of x

3. If $A, B, A + I, A + B$ are idempotent matrices (where A and B are square matrices of same order & I is identity matrix of same order), then value of $|AB - BA|$ (where $|C|$ denotes determinant of matrix C) is -

- (A) 0 (B) 1
(C) 2 (D) cannot be determined

4. If A is nilpotent matrix of index 2, then $A(I_2 + A)^{51}$ is equal to (where I_2 is the identity matrix of order 2) -

- (A) A^{51} (B) I_2 (C) Null matrix (D) A

5. If A and B are two idempotent matrices of same order and $A + B$ is idempotent, then -

- (A) $AB + BA = O$ (B) $AB = I$ (C) $A = B$ (D) none of these

6. If A and B are square matrices such that $AB = B$ and $BA = A$, then which of the following is/are always true ?

- (A) A is an idempotent matrix (B) B is an involutory matrix
(C) $A^6 + B^6 = A^8 + B^8$ (D) $A^2 + B^2 = A + I$

7. If $\begin{bmatrix} a & b \\ c & 1-a \end{bmatrix}$ is an idempotent matrix. Find the value of $f(a)$, where $f(x) = x - x^2$, when $bc = 1/4$.

Hence otherwise evaluate a .

8. If the matrix A is involutory, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and

$$\frac{1}{2}(I + A) \cdot \frac{1}{2}(I - A) = O.$$

12. THE TRANSPOSE OF A MATRIX : (Changing rows & columns)

Let A be any matrix of order $m \times n$. Then A^T or $A' = [a_{ji}]$ for $1 \leq i \leq m$ & $1 \leq j \leq n$ of order $n \times m$

Properties of transpose :

If A^T & B^T denote the transpose of A and B,

(a) $(A+B)^T = A^T + B^T$; note that A & B have the same order.

(b) $(AB)^T = B^T A^T$ (Reversal law) A & B are conformable for matrix product AB

Note : In general : $(A_1 \cdot A_2 \cdot \dots \cdot A_n)^T = A_n^T \cdot \dots \cdot A_2^T \cdot A_1^T$ (reversal law for transpose)

(c) $(A^T)^T = A$

(d) $(kA)^T = kA^T$, k is a scalar.

Illustration 10 : If A and B are matrices of order $m \times n$ and $n \times m$ respectively (where $m \neq n$), then order of matrix $B^T(A^T)^T$ is -

- (A) $m \times n$ (B) $m \times m$ (C) $n \times n$ (D) Not defined

Solution : Order of B is $n \times m$ so order of B^T will be $m \times n$

Now $(A^T)^T = A$ & its order is $m \times n$. For the multiplication $B^T(A^T)^T$

Number of columns in prefactor $\neq$ Number of rows in post factor.

Hence this multiplication is not defined.

Ans. (D)

13. ORTHOGONAL MATRIX

A square matrix is said to be orthogonal matrix if $AA^T = I$

Note :

(i) The determinant value of orthogonal matrix is either 1 or -1.

(ii) Let $A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$

$$AA^T = \begin{bmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ b_1a_1 + b_2a_2 + b_3a_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ c_1a_1 + c_2a_2 + c_3a_3 & c_1b_1 + c_2b_2 + c_3b_3 & c_1^2 + c_2^2 + c_3^2 \end{bmatrix}$$

If $AA^T = I$, then

$$\sum_{i=1}^3 a_i^2 = \sum_{i=1}^3 b_i^2 = \sum_{i=1}^3 c_i^2 = 1 \text{ and } \sum_{i=1}^3 a_i b_i = \sum_{i=1}^3 b_i c_i = \sum_{i=1}^3 c_i a_i = 0$$

Illustration 11 : Determine the values of α, β, γ , when $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Solution : Let $A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$

$$\therefore A' = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$$

But given A is orthogonal.

$$\therefore AA^T = I$$

$$\Rightarrow \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we have

$$4\beta^2 + \gamma^2 = 1 \quad \text{.....(i)}$$

$$2\beta^2 - \gamma^2 = 0 \quad \text{.....(ii)}$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \quad \text{.....(iii)}$$

From (i) and (ii), $6\beta^2 = 1 \quad \therefore \beta^2 = \frac{1}{6} \quad \text{and} \quad \gamma^2 = \frac{1}{3}$

From (iii) $\alpha^2 = 1 - \beta^2 - \gamma^2 = 1 - \frac{1}{6} - \frac{1}{3} = \frac{1}{2}$

Hence, $\alpha = \pm \frac{1}{\sqrt{2}}, \beta = \pm \frac{1}{\sqrt{6}} \quad \text{and} \quad \gamma = \pm \frac{1}{\sqrt{3}}$

Ans.

14. SYMMETRIC & SKEW SYMMETRIC MATRIX :

(a) Symmetric matrix :

A square matrix $A = [a_{ij}]$ is said to be, symmetric if, $a_{ij} = a_{ji} \forall i \& j$ (conjugate elements are equal).
Hence for symmetric matrix $A = A^T$.

Note : Max. number of distinct entries in any symmetric matrix of order n is $\frac{n(n+1)}{2}$.

(b) Skew symmetric matrix :

Square matrix $A = [a_{ij}]$ is said to be skew symmetric if $a_{ij} = -a_{ji} \forall i \& j$ (the pair of conjugate elements are additive inverse of each other). For a skew symmetric matrix $A = -A^T$.

Note :

- (i) If A is skew symmetric, then $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0 \forall i$. Thus the diagonal elements of a skew square matrix are all zero, but not the converse.
- (ii) The determinant value of odd order skew symmetric matrix is zero.

(c) Properties of symmetric & skew symmetric matrix :

- (i) A is symmetric if $A^T = A$ & A is skew symmetric if $A^T = -A$
- (ii) Let A be any square matrix then, $A + A^T$ is a symmetric matrix & $A - A^T$ is a skew symmetric matrix.
- (iii) The sum of two symmetric matrix is a symmetric matrix and the sum of two skew symmetric matrix is a skew symmetric matrix.
- (iv) If A & B are symmetric matrices then,
 - (1) $AB + BA$ is a symmetric matrix
 - (2) $AB - BA$ is a skew symmetric matrix.
- (v) Every square matrix can be uniquely expressed as a sum or difference of a symmetric and a skew symmetric matrix.

$$A = \underbrace{\frac{1}{2} (A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2} (A - A^T)}_{\text{skew symmetric}} \text{ and } A = \frac{1}{2} (A^T + A) - \frac{1}{2} (A^T - A)$$

Illustration 12: If A is symmetric as well as skew symmetric matrix, then A is -

- (A) diagonal matrix (B) null matrix (C) triangular matrix (D) none of these

Solution :

Let $A = [a_{ij}]$ Since A is skew symmetric $a_{ij} = -a_{ji}$
for $i = j$, $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$
for $i \neq j$, $a_{ij} = -a_{ji}$ [$\because A$ is skew symmetric] & $a_{ij} = a_{ji}$ [$\because A$ is symmetric]
 $\therefore a_{ij} = 0$ for all $i \neq j$
so, $a_{ij} = 0$ for all 'i' and 'j' i.e. A is null matrix.

Ans. (A,B,C)

Do yourself - 6 :

- If $A = \begin{bmatrix} -2 & -1 & 1 \\ -1 & 7 & 4 \\ 1 & -x & -3 \end{bmatrix}$ be symmetric matrix then find the value of x .
- Express matrix $A = \begin{bmatrix} 2 & 5 & 7 \\ 9 & -7 & 2 \\ 1 & -1 & 0 \end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.
- If A is a matrix of order 3×3 such that $A^2 + A = A^T$, then which of the following is wrong -
 (A) $|A| = 0$ (B) $|A| \neq 0$
 (C) A^2 is a skew symmetric matrix (D) A is singular matrix
- Let $A = \begin{bmatrix} 0 & \tan \theta & 2 \\ -\tan \theta & 0 & \sec \theta \\ -2 & -\sec \theta & 0 \end{bmatrix}$, where $\theta \neq (2n+1)\frac{\pi}{2}$, ($n \in \mathbb{I}$) & $B = \sum_{r=1}^5 A^{(2r-1)}$ then which of the following statement is correct
 (A) A is skew symmetric matrix (B) B is skew symmetric matrix
 (C) $(A+B)$ is skew symmetric matrix (D) $(A+B)$ is symmetric matrix
- If A, B are symmetric matrices of the same order then $AB - BA$ is -
 (A) a symmetric matrix (B) a skew symmetric matrix
 (C) unit matrix (D) a null matrix
- Let A and B be 3×3 matrices of real number where A is symmetric, B is skew symmetric and $(A+B)(A-B) = (A-B)(A+B)$ if $(AB)^T = (-1)^k AB$, then possible value of k is -
 (A) 3 (B) 4 (C) 6 (D) 2
- Which of the following matrices $A = [a_{ij}]_{n \times n}$ represents a symmetric matrix ?
 (A) $a_{ij} = i - j + \sin(i+j) \cdot \sin(i-j)$ (B) $a_{ij} = |i-j| + \sin(|i+j|) \sin(i-j)$
 (C) $a_{ij} = \tan^{-1}(\tan(i+j)) + i - j$ (D) $a_{ij} = \cos(i-j) \cdot \cos(i+j) - (i-j)^2$
- Let A is symmetric and B is a skew symmetric matrix, such that $A - B = \begin{bmatrix} 2 & 5 \\ 7 & 6 \end{bmatrix}$, then $\frac{(|A|)^2}{96}$ is
- If A and B are symmetric matrices, then prove that ABA is symmetric as well.
- Let $A = \begin{pmatrix} 0 & \sin \alpha & \sin \alpha \sin \beta \\ -\sin \alpha & 0 & \cos \alpha \cos \beta \\ -\sin \alpha \sin \beta & -\cos \alpha \cos \beta & 0 \end{pmatrix}$, then prove that $|A|$ is independent of α and β .
- Let $A = a_{ij}$ be a matrix of order 3 where $a_{ij} = \begin{cases} x & \text{if } i = j, \quad x \in \mathbb{R} \\ 1 & \text{if } |i-j|=1 \\ 0 & \text{otherwise} \end{cases}$, then which of the following hold(s) good ?
 (A) for $x = 2$, A is a diagonal matrix.
 (B) A is a symmetric matrix
 (C) for $x = 2$, $\det A$ has the value equal to 6
 (D) Let $f(x) = \det A$, then the function $f(x)$ has both local maxima and local minima.

15. ADJOINT OF A SQUARE MATRIX :

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A then the adjoint of A , denoted by $\text{adj}A$, is defined as the transpose of the cofactor matrix.

$$\text{Then, } \text{adj}A = [C_{ij}]^T \Rightarrow \text{adj}A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{23} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$$

Theorem : $A (\text{adj. } A) = (\text{adj. } A) . A = |A| I_n$.

$$\text{Proof : } A.(\text{adj } A) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} C_{11} & C_{21} & C_{31} \\ C_{12} & C_{22} & C_{32} \\ C_{13} & C_{23} & C_{33} \end{pmatrix}$$

$$\begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix} = |A| \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow A.(\text{Adj. } A) = |A| I$$

(whatever may be the value only $|A|$ will come out as a common element)

If $|A| \neq 0$, then $\frac{A.(\text{adj.}A)}{|A|} = I = \text{unit matrix of the same order as that of } A$

Properties of adjoint matrix :

If A be a square matrix of order n , then

- (i) $|\text{adj } A| = |A|^{n-1}$
- (ii) $\text{adj}(\text{adj } A) = |A|^{n-2} A$, where $|A| \neq 0$
- (iii) $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$, where $|A| \neq 0$
- (iv) $\text{adj}(AB) = (\text{adj } B) (\text{adj } A)$
- (v) $\text{adj}(KA) = K^{n-1} (\text{adj } A)$, K is a scalar
- (vi) $\text{adj } A^T = (\text{adj } A)^T$

Method to find adjoint of a 2×2 square matrix, directly :

Let A be a 2×2 square matrix. In order to find the adjoint simply interchange the diagonal elements and reverse the sign of off diagonal elements (rest of the elements).

$$\text{e.g. If } A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow \text{adj}A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$$

Illustration 13: If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then $\text{adj } A$ is equal to -

(A) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$

(B) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$

(D) none of these

Solution :

$$\text{adj. } A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$$

Ans. (A)

Illustration 14: If $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$, then $\text{adj}(\text{adj } A)$ is equal to -

(A) $8 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(B) $16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(C) $64 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(D) none of these

Solution :

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 8$$

$$\text{Now } \text{adj}(\text{adj } A) = |A|^{3-2} A$$

$$= 8 \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Ans. (B)

Do yourself - 7 :

- For any 2×2 matrix, if $A(\text{Adj}A) = \begin{bmatrix} 15 & 0 \\ 0 & 15 \end{bmatrix}$, then $|A|$ is equal -
 (A) 20 (B) 625 (C) 15 (D) 0
- Which of the following is/are incorrect ?
 (A) Adjoint of a symmetric matrix is skew symmetric matrix.
 (B) Adjoint of a diagonal matrix is a diagonal matrix.
 (C) $A(\text{Adj}A) = (\text{Adj}A)A = |A|I$
 (D) Adjoint of a unit matrix is a diagonal matrix
- If A be a square matrix of the order 5 and $B = \text{Adj}(A)$ then find $\text{Adj}(5A)$.
- If A be a square matrix of order 4 and $|A| = 3$ then find $\text{adj}(\text{adj}A)$.
- If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 1 & 3 \end{bmatrix}$, then $\text{tr}(A^3 \text{adj}(A^2))$, is
 (where $\text{tr}(A)$ and $\text{adj}(A)$ denote trace and adjoint of matrix A) -
 (A) 6 (B) 24 (C) 36 (D) 12
- Let A and B are two non-singular square matrices of order n with real entries such that $\text{adj}A = \text{adj}B$, then which of the following is necessarily true -
 (A) $A = B$ if n is even (B) $A = -B$ if n is even
 (C) $A = -B$ if n is odd (D) $A = B$ if n is odd
- Let A be square matrix of order 3. If $\det A = 2$, then value of $\det(\text{adj} A^3)$ is equal to
 (Here $\det A$ denotes determinant of matrix A)
 (A) 8 (B) 28 (C) 32 (D) 64
- If A is a square matrix of order n with $|A| = 9$ and $|\text{adj adj adj}A| = 3^{16}$, then number of elements in A , is -
 (A) 9 (B) 4 (C) 5 (D) 2
- If A and B are non-singular matrices of order three such that $\text{adj}(AB) = \begin{bmatrix} 1 & 1 & 1 \\ 1 & p & 1 \\ 1 & 1 & p \end{bmatrix}$ and
 $|B^2 \text{adj}(A)| = p^2 - 3$, then p is
- $A_{3 \times 3}$ is a matrix such that $|A| = a$, $B = (\text{adj} A)$ such that $|B| = b$. Find the value of $(ab^2 + a^2b + 1)S$
 where $\frac{1}{2}S = \frac{a}{b} + \frac{a^2}{b^3} + \frac{a^3}{b^5} + \dots$ up to ∞ , and $a = 3$.

16. INVERSE OF A MATRIX (Reciprocal Matrix) :

A square matrix A said to be invertible if and only if it is non-singular (i.e. $|A| \neq 0$) and there exists a matrix B such that, $AB = I = BA$.

B is called the **inverse** (reciprocal) of A and is denoted by A^{-1} . Thus

$$A^{-1} = B \Leftrightarrow AB = I = BA$$

We have, $A \cdot (\text{adj } A) = |A| I_n$

$$A^{-1} \cdot A(\text{adj } A) = A^{-1} |A| I_n$$

$$I_n (\text{adj } A) = A^{-1} |A| I_n$$

$$\therefore A^{-1} = \frac{(\text{adj } A)}{|A|}$$

Note : The necessary and sufficient condition for a square matrix A to be invertible is that $|A| \neq 0$

Properties of inverse :

(i) If A & B are invertible matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Note: If $A_1, A_2, \dots, A_n$ are all invertible square matrices of order n

$$\text{then } (A_1 A_2 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_2^{-1} A_1^{-1}$$

(ii) If A be an invertible matrix, then A^T is also invertible & $(A^T)^{-1} = (A^{-1})^T$.

(iii) If A is invertible, (a) $(A^{-1})^{-1} = A$ (b) $(A^k)^{-1} = (A^{-1})^k = A^{-k}$; $k \in \mathbb{N}$

(iv) If A is non-singular matrix, then $|A^{-1}| = |A|^{-1}$

(v) If idempotent matrix is invertible then its inverse will be identity matrix.

(vi) A nilpotent matrix will not be invertible because its determinant value is zero.

(vii) Orthogonal matrix A is always invertible and $A^{-1} = A^T$.

(viii) $A = A^{-1}$ for an involutory matrix.

Cancellation law : Let A, B, C be square matrices of the same order 'n'.

If A is a non-singular matrix, then

(a) $AB = AC \Rightarrow B = C$ (Left cancellation law)

(b) $BA = CA \Rightarrow B = C$ (Right cancellation law)

Note that these cancellation laws hold only if the matrix ' A ' is **non-singular** (i.e. $|A| \neq 0$).

Illustration 15 : Prove that if A is non-singular matrix such that A is symmetric then A^{-1} is also symmetric.

Solution : $A^T = A$ [$\because A$ is a symmetric matrix]

$$(A^T)^{-1} = A^{-1} \text{ [since } A \text{ is non-singular matrix]}$$

$$\Rightarrow (A^{-1})^T = A^{-1} \text{ Hence proved}$$

Illustration 16 : $\begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1}$ is equal to -

- (A) $\begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$ (B) $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ (C) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (D) none of these

Solution :

$$\begin{aligned} \begin{bmatrix} 1 & \tan \theta/2 \\ -\tan \theta/2 & 1 \end{bmatrix}^{-1} &= \frac{1}{\sec^2 \theta/2} \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \\ \therefore \text{Product} &= \frac{1}{\sec^2 \theta/2} \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \theta/2 \\ \tan \theta/2 & 1 \end{bmatrix} \\ &= \frac{1}{\sec^2 \theta/2} \begin{bmatrix} 1 - \tan^2 \theta/2 & -2 \tan \theta/2 \\ 2 \tan \theta/2 & 1 - \tan^2 \theta/2 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta/2 & \sin^2 \theta/2 & -2 \sin \theta/2 \cos \theta/2 \\ 2 \sin \theta/2 & \cos^2 \theta/2 & \cos^2 \theta/2 - \sin^2 \theta/2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \end{aligned}$$

Ans. (C)

Illustration 17 : If $A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $M = AB$, then M^{-1} is equal to-

- (A) $\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$ (C) $\begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$ (D) $\begin{bmatrix} 1/3 & -1/3 \\ -1/3 & 1/6 \end{bmatrix}$

Solution :

$$M = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 2 \end{bmatrix}$$

$$|M| = 6, \text{adj } M = \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$$

$$\therefore M^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$$

Ans. (C)

Do yourself - 8 :

- If A' is a square matrix such that $A^2 = I$ then A^{-1} is equal to -
 (A) $A + I$ (B) A (C) 0 (D) $2A$
- If A' is an orthogonal matrix, then A^{-1} equals -
 (A) A (B) A^T (C) A^2 (D) none of these
- If $A = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$, then $(A^{-1})^3$ is equal to -
 (A) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & 27 \end{bmatrix}$ (B) $\frac{1}{27} \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}$ (C) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & -27 \end{bmatrix}$ (D) $\frac{1}{27} \begin{bmatrix} -1 & -26 \\ 0 & -27 \end{bmatrix}$
- If A is a non-zero idempotent matrix satisfying $\left(I - \frac{2}{5}A\right)^{-1} = I - \alpha A$ where I is the unit matrix of the same order as that of A then the absolute value of 3α is
- Let A , B and $A + B$ are non-singular matrices of order 3×3 satisfying
 $A^{-1} + B^{-1} = (A + B)^{-1}$ and $|AB^{-1}| \in \mathbb{R}$ then value of $\frac{|A|}{|B|}$ is
- Let P , Q , R be invertible matrices of second order such that $A = PQ^{-1}$, $B = QR^{-1}$, $C = RP^{-1}$, then $\det(ABC + BCA + CAB)$ is
- An invertible matrix A of order 3 satisfies the relation $A = A^{-1} + 2I$, (where I denotes identity matrix). The value of $|A - I| \cdot |A + I| \cdot |A - 2I|$ is
- If A is a square matrix such that $A^5 = O$ and $A^n \neq O$ for $n = 1, 2, 3, 4$, then $(I - A)^{-1}$ is equal to (where O is a null matrix)-
 (A) $I + A$ (B) $I + A + A^2$ (C) $I + A^{-1}$ (D) $I + A + A^2 + A^3 + A^4$
- If A and B are non-singular symmetric matrices of the same order such that $AB = BA$ and $P = (AB^{-1})^T$ & $Q = (A^{-1}B)^T$ then $(PQ)^{-1}$ is equal to -
 (A) $AB^{-1}AB$ (B) $A^{-1}B^{-1}AB$ (C) $A^{-1}BA^{-1}B$ (D) A^2B^2
- Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of matrix A , then α is -
 (A) -2 (B) -1 (C) 2 (D) 5

11. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then let us define a function $f(x) = \det. (A^T A^{-1})$ then which of the following can not be the value of $\underbrace{f(f(f(\dots\dots\dots f(x))))}_{n \text{ times}}$ is ($n \geq 2$)
- (A) $f^n(x)$ (B) 1 (C) $f^{n-1}(x)$ (D) $n f(x)$
12. If A and B are 3×3 matrices and $|A| \neq 0$, then which of the following are true?
- (A) $|AB| = 0 \Rightarrow |B| = 0$ (B) $|AB| = 0 \Rightarrow B = 0$
 (C) $|A^{-1}| = |A|^{-1}$ (D) $|A + A| = 2|A|$
13. Let $A^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$, then -
- (A) $7|A| = \frac{1}{2}$ (B) $|\text{adj } A| = \frac{1}{196}$
 (C) $\text{trace}(\text{adj } A) = -\frac{1}{7}$ (D) Matrix A is a symmetric matrix
14. If $A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 4 & 5 & 3 \end{bmatrix}$, then which of the following is(are) true ?
- (trace of A denotes sum of principal diagonal elements of A)
- (A) A is invertible (B) $\text{trace}(\text{adj}(\text{adj}(A))) = 144$
 (C) $\text{trace}(\text{adj}(\text{adj}(A))) = 8$ (D) $|\text{adj } A|$ is less than 400
15. For the matrix $A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}$ find A^{-2} .
16. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix}$; $C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ and $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, then solve the following matrix equation.
- (a) $AX = B - I$ (b) $(B - I)X = IC$ (c) $CX = A$

17. MATRIX POLYNOMIAL :

If $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_n x^0$, then we define a matrix polynomial

$$f(A) = a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I^n.$$

where A is the given square matrix. If $f(A)$ is the null matrix, then A is called the zero or root of the polynomial $f(x)$

18. CHARACTERISTIC EQUATION :

Let A be a square matrix. Then the polynomial $|A - xI|$ is called as characteristic polynomial of A & the equation $|A - xI| = 0$ is called as characteristic equation of A . After solving the characteristic polynomial the values of ' x ' are said to be characteristic roots of the polynomial.

Note : (i) Sum of the roots of the characteristic equation is equal to trace of the matrix.

(ii) Product of the roots of the characteristic equation is equal to the determinant value.

(iii) The degree of characteristic equation is same as the order of the matrix.

Illustration 18 : If $f(x) = x^2 - 3x + 3$ and $A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$ be a square matrix then prove that $f(A) = \mathbf{O}$. Hence find A^4 .

Solution : $A^2 = A.A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix}$

$$\text{Hence } A^2 - 3A + 3I = \begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix} - 3 \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{O}$$

Aliter : $\because |A - XI| = 0 \Rightarrow \begin{vmatrix} 2-x & 1 \\ -1 & 1-x \end{vmatrix} = 0$

$$\Rightarrow (2-x)(1-x) + 1 = 0 \Rightarrow x^2 - 3x + 3 = 0 \quad (\text{characteristic polynomial})$$

by Cayley-Hamilton Theorem $A^2 - 3A + 3I = \mathbf{O}$. Hence proved.

Now $A^2 = 3A - 3I$

squaring on both the sides

$$A^4 = 9(A^2 - 2A + I)$$

$$= 9 \left(\begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix} - 2 \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 9 \begin{bmatrix} 3-4+1 & 3-2 \\ -3+2 & -2+1 \end{bmatrix}$$

$$= 9 \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 9 \\ -9 & -9 \end{bmatrix}$$

19. CAYLEY - HAMILTON THEOREM :

Every square matrix A satisfy its characteristic equation

i.e. $a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0$ is the characteristic equation of A , then

$$a_0A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = \mathbf{O}$$

Note : This theorem is helpful to find the inverse of any non-singular square matrix.

$$\text{i.e. } a_0A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = \mathbf{O}$$

On multiplying by A^{-1} on both the sides of above equation, we get

$$A^{-1} = -\frac{1}{a_n} (a_0A^{n-1} + a_1A^{n-2} + \dots + a_{n-1}I)$$

Illustration 19: If $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$, show that $5A^{-1} = A^2 + A - 5I$

Solution : We have the characteristic equation of A.

$$|A - xI| = 0$$

$$\text{i.e. } \begin{vmatrix} 1-x & 2 & 0 \\ 2 & -1-x & 0 \\ 0 & 0 & -1-x \end{vmatrix} = 0$$

$$\text{i.e. } x^3 + x^2 - 5x - 5 = 0$$

Using Cayley–Hamilton theorem

$$A^3 + A^2 - 5A - 5I = \mathbf{O} \quad \Rightarrow \quad 5I = A^3 + A^2 - 5A$$

Multiplying by A^{-1} , we get $5A^{-1} = A^2 + A - 5I$

Do yourself - 9 :

1. Determine the characteristic roots of the matrix A. Hence find the trace and determinant value of A.

Where $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and also prove that $A^3 - 18A^2 + 45A = \mathbf{O}$.

2. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ is zero divisor of $f(x) = x^2 - 4x - 5$, then A^3 is equal to

(A) $\begin{bmatrix} 41 & 42 & 42 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{bmatrix}$

(B) $\begin{bmatrix} 42 & 41 & 42 \\ 41 & 42 & 41 \\ 44 & 41 & 42 \end{bmatrix}$

(C) $\begin{bmatrix} 43 & 42 & 41 \\ 44 & 41 & 42 \\ 42 & 43 & 43 \end{bmatrix}$

(D) $\begin{bmatrix} 41 & 42 & 41 \\ 42 & 41 & 42 \\ 42 & 42 & 41 \end{bmatrix}$

3. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ & all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$, then the value of $(x + y + z)$ where I is a unit matrix of order 3, is -

(A) 0

(B) 1

(C) 2

(D) 3

4. If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.

20. SYSTEM OF EQUATION & CRITERIA FOR CONSISTENCY

Gauss - Jordan method :

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

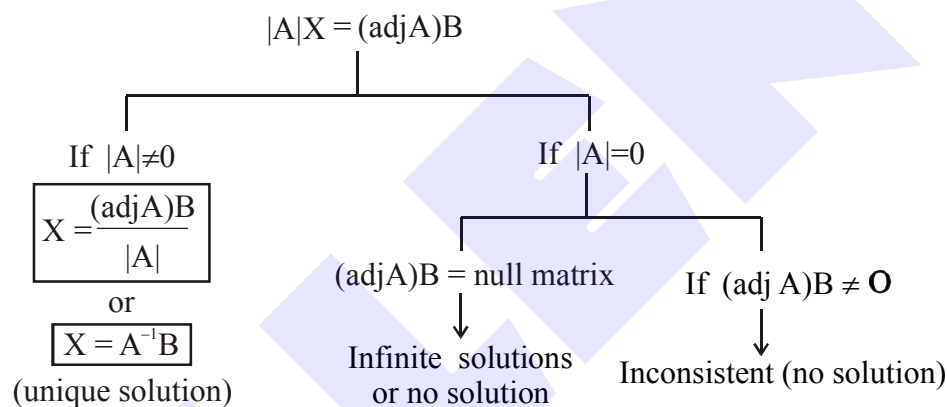
$$a_3x + b_3y + c_3z = d_3$$

$$\Rightarrow \begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$\Rightarrow AX = B \quad \dots(i)$$

Multiplying adjA on both the sides of (i)

$$\Rightarrow (\text{adj}A)AX = (\text{adj}A)B \Rightarrow |A|X = (\text{adj}A)B$$



$$x + y + z = 6$$

Illustration 20: Solve the system $x - y + z = 2$ using matrix method.

$$2x + y - z = 1$$

Solution :

$$\text{Let } A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ \& } B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

Then the system is $AX = B$. $|A| = 6$, hence A is non singular,

$$\text{Cofactor } A = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix}$$

$$X = A^{-1} B = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \quad \text{i.e.} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3$$

Ans.

Do yourself - 10 :

- The system of equations $x + 2y - 3z = 1$, $x - y + 4z = 0$, $2x + y + z = 1$ has -
 (A) only two solutions (B) only one solution
 (C) no solution (D) infinitely many solutions
- The system of equations $x + y + z = 8$, $x - y + 2z = 6$, $3x + 5y - 7z = 14$ has-
 (A) Unique solution (B) infinite number of solutions
 (C) no solution (D) none of these
- If the system of linear equations $x + 2ay + az = 0$, $x + 3by + bz = 0$, $x + 4cy + cz = 0$, has a non-trivial solution & $a, b, c > 0$ then the minimum value of $\left(\frac{a^3 + c^3}{b^3}\right)$ is
- Square matrices B of second order are formed using some or all the possible values of k, when system of linear equations $kx + 4y + z = 0$, $4x + ky + 2z = 0$ & $2x + 2y + z = 0$, has infinite solution, then number of singular matrices B is
- Consider the system of equations $x - 2y + 3z = -1$, $x - 3y + 4z = 1$ and $x + y + 2z = k$ has exactly three solutions, then -
 (A) $k = 3$ (B) $k = 0$ (C) $k = -9$ (D) $k = \phi$